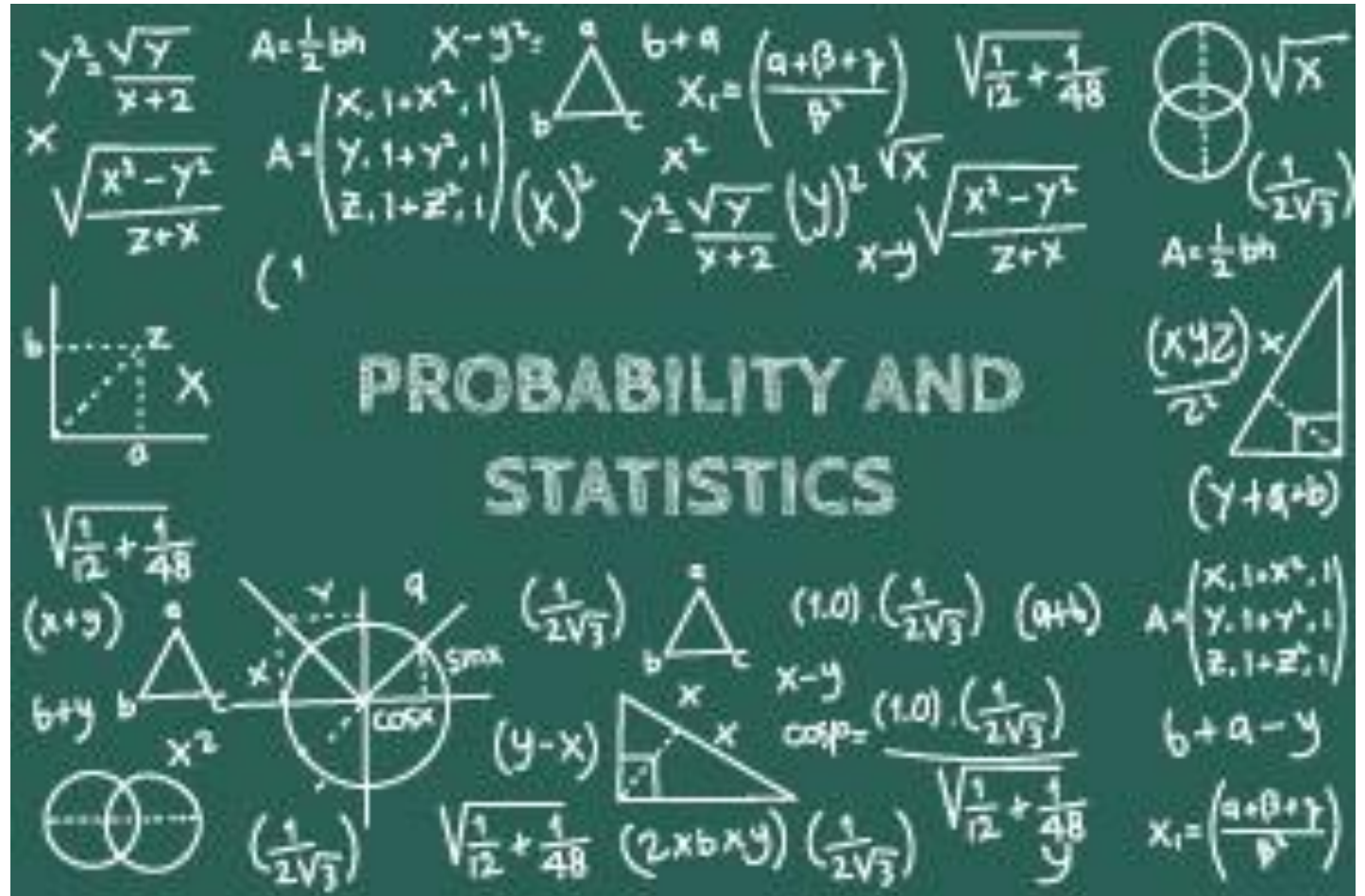


Lec # 07
ES111

Independent, Product, Total Probab.

Book Reading: Sec. 2.6, 2.7

@M. Tahir Naseem



Discussion Plan

1. **Independent** Events

2. **Product** Rule

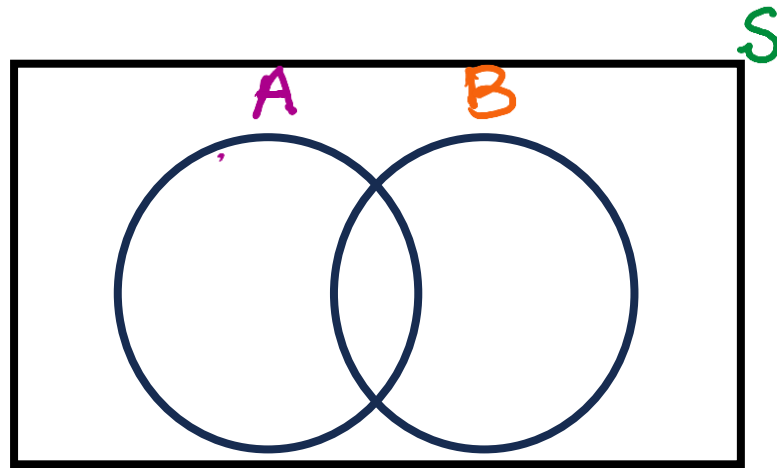
3. **Total** Probability

(Sec. 2.6, 2.7)

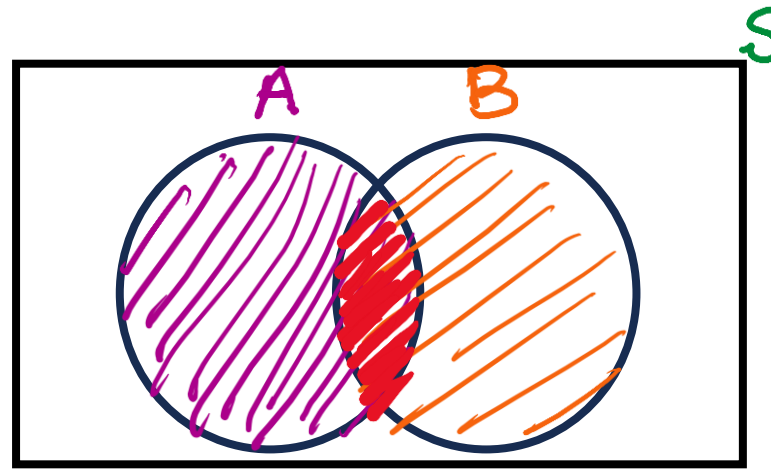


conditional Probability

conditional Probability



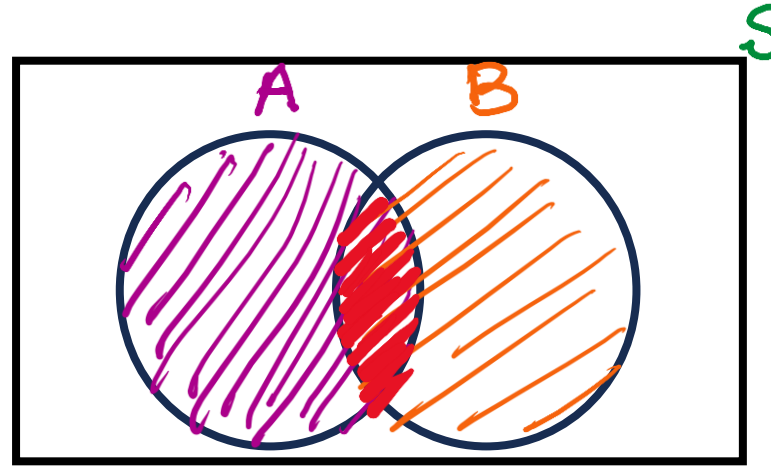
conditional Probability



we want to find

$$P(A/B) =$$

conditional Probability



we want to find

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$



A die is loaded in such a way that an **even number is twice** as likely to occur **as an odd number**. **What is the probability** of getting **perfect square** given the die resulted in a number greater than 3.



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↓
"condition"

$$S = \{1, 2, 3, 4, 5, 6\}$$

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$$S = \{1, 2, 3, 4, 5, 6\}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A/B) = \frac{2/9}{5/9} \quad \underline{\underline{\text{Ans}}}$$

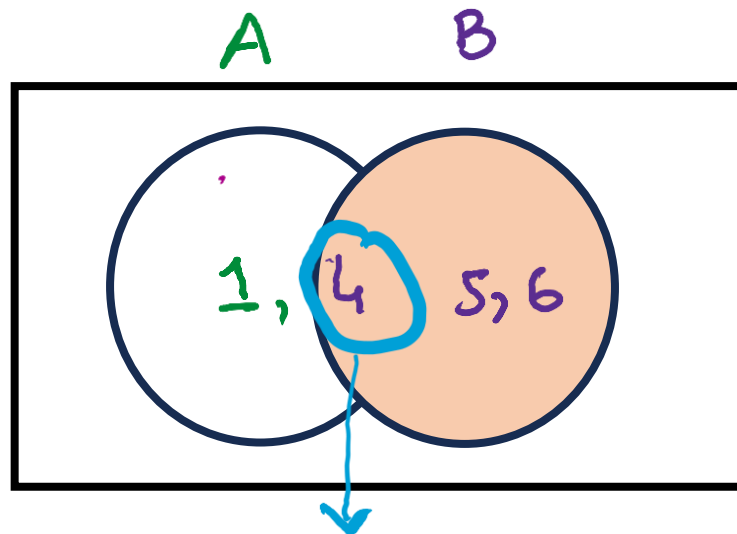
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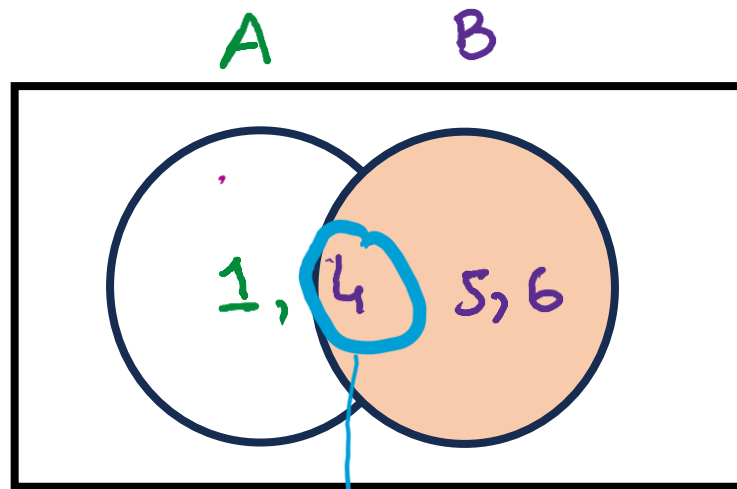
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$$P(A/B) = \frac{2/9}{5/9} \text{ Ans}$$

$$A = \{1, 4\}$$

$$B = \{4, 5, 6\}$$

Independent Events

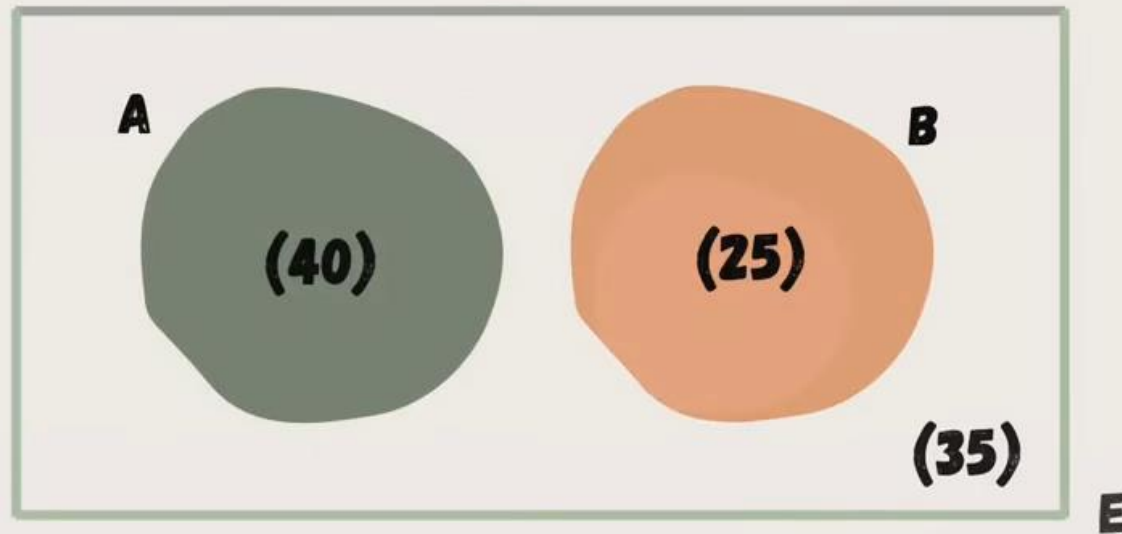
Independent Events

MUTUALLY EXCLUSIVE EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
A student plays
Basketball

EVENT B:
A student plays
Cricket



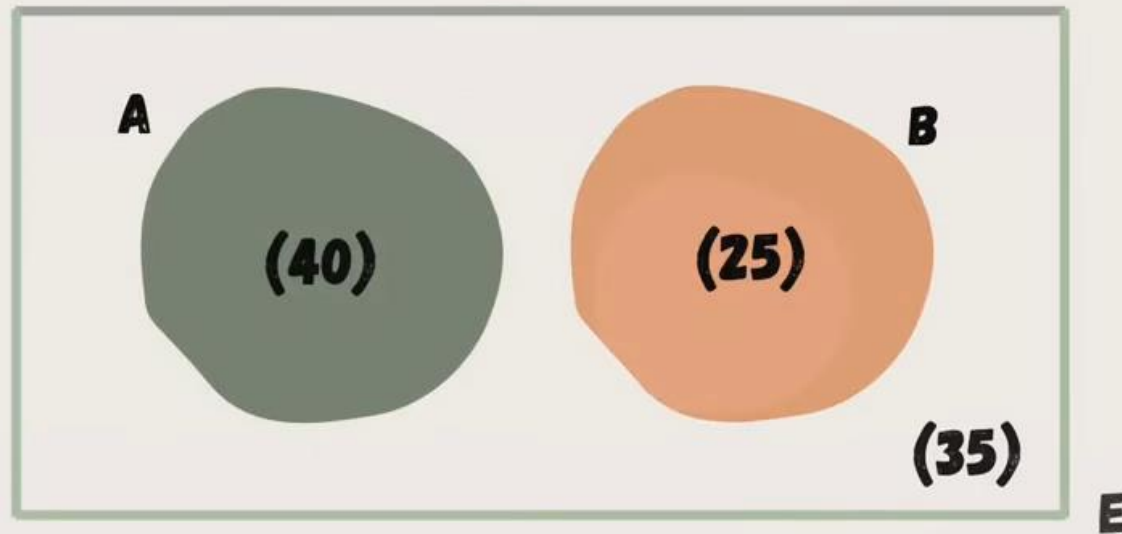
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$$P(A) = \frac{40}{100} = 0.4$$

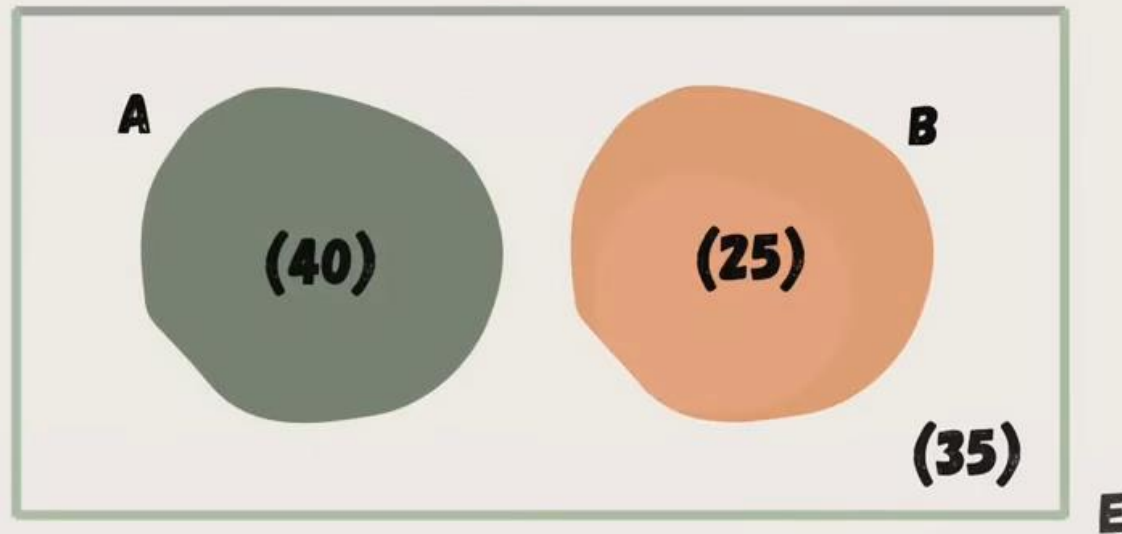
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$$P(A) = \frac{40}{100} = 0.4$$

$$P(B) = \frac{25}{100} = 0.25$$

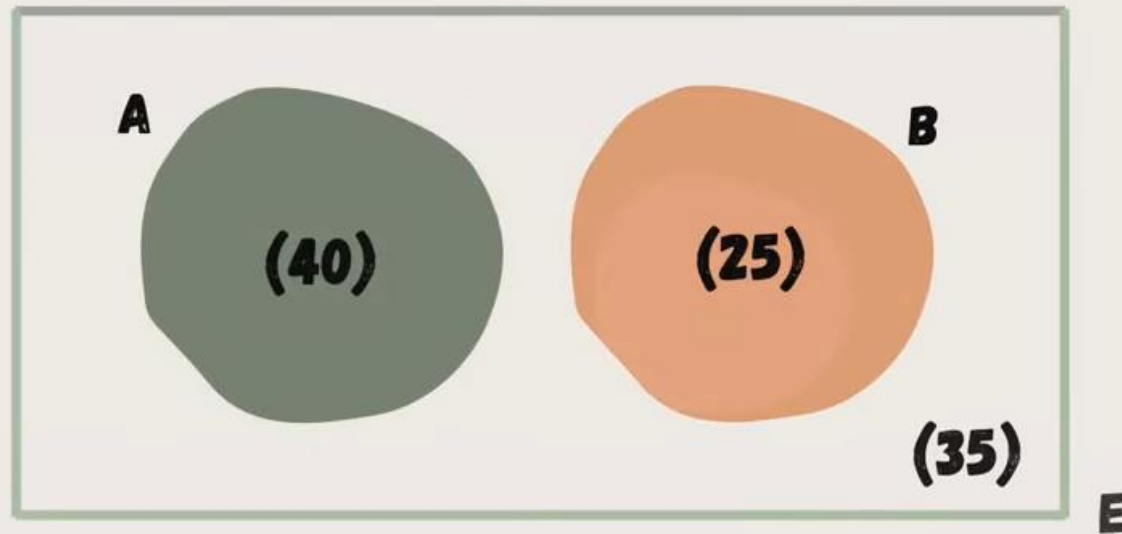
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$$P(A \cap B) = 0$$

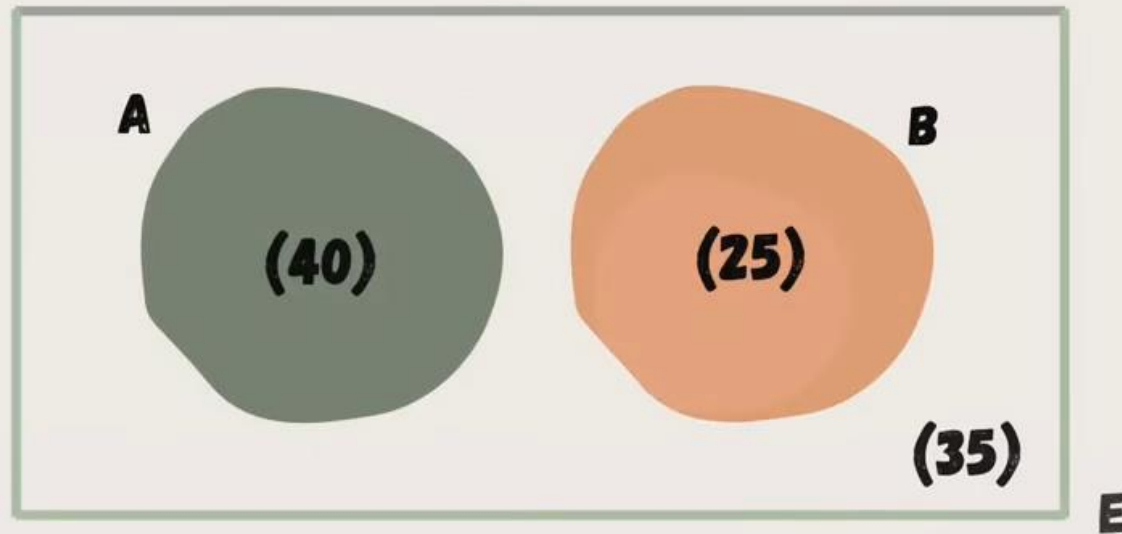
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$$P(A) = \frac{40}{100} = 0.4$$

$$P(B) = \frac{25}{100} = 0.25$$

$$P(A \cap B) = 0$$

$$P(A \cap B) = \{\emptyset\}$$

Mutually Exclusive

Independent Events

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed



EVENT A:

A student plays
Basketball

EVENT B:

A student studies
Modern History



Independent Events

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
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$$P(A) = \frac{40}{100} = 0.4$$

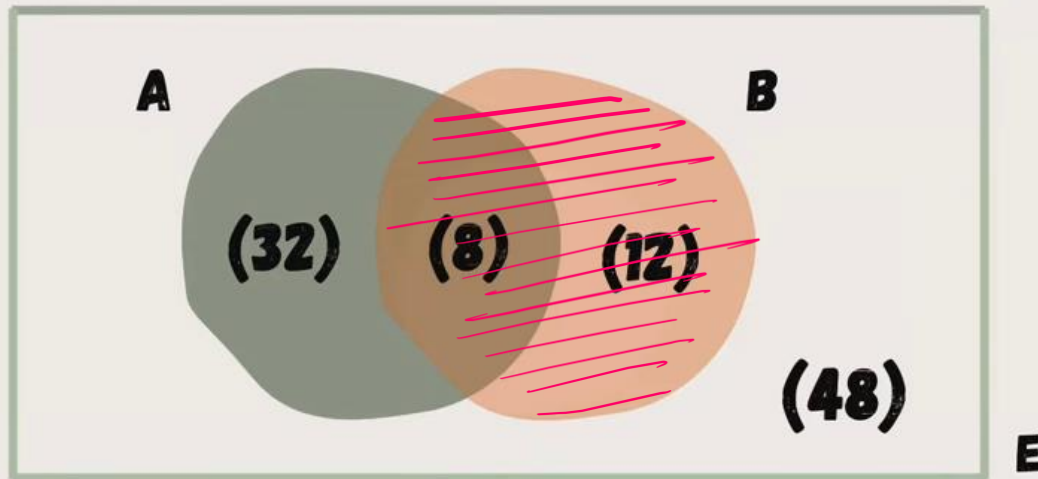
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$$P(A/B) =$$

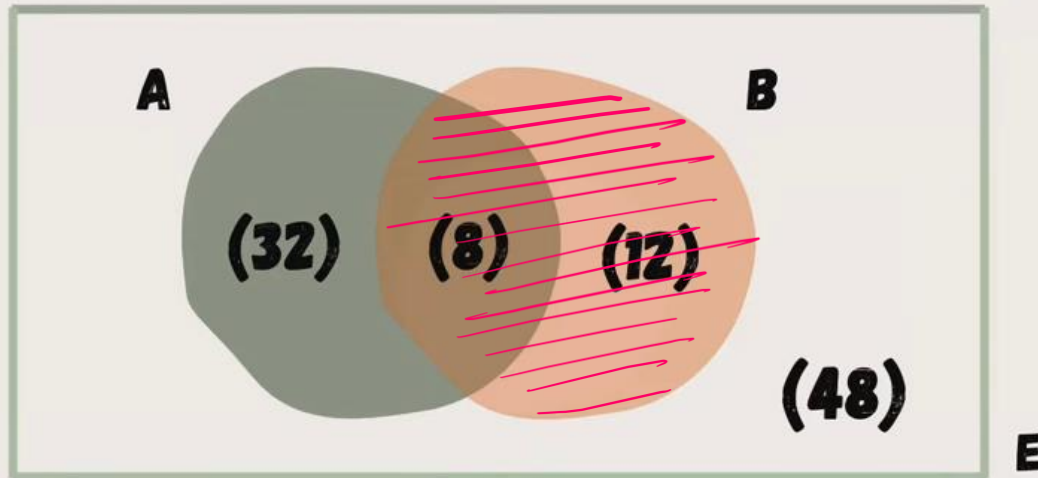
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$$P(A) = \frac{40}{100} = 0.4$$

$$P(A|B) = \frac{8}{20} = 0.4$$

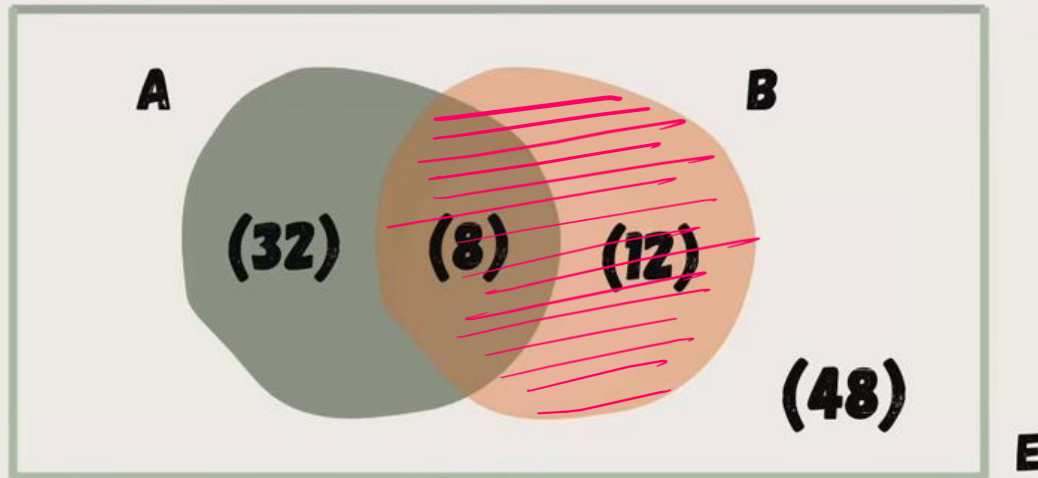
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$$P(A) = \frac{40}{100} = 0.4$$

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$$= \frac{P(A \cap B)}{P(B)}$$

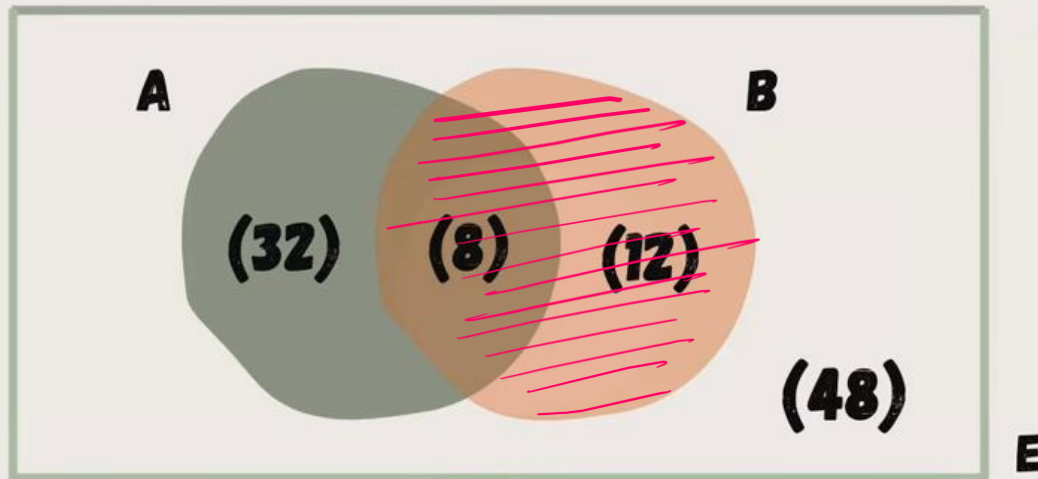
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$$= \frac{P(A \cap B)}{P(B)} = \frac{8/100}{20/100}$$

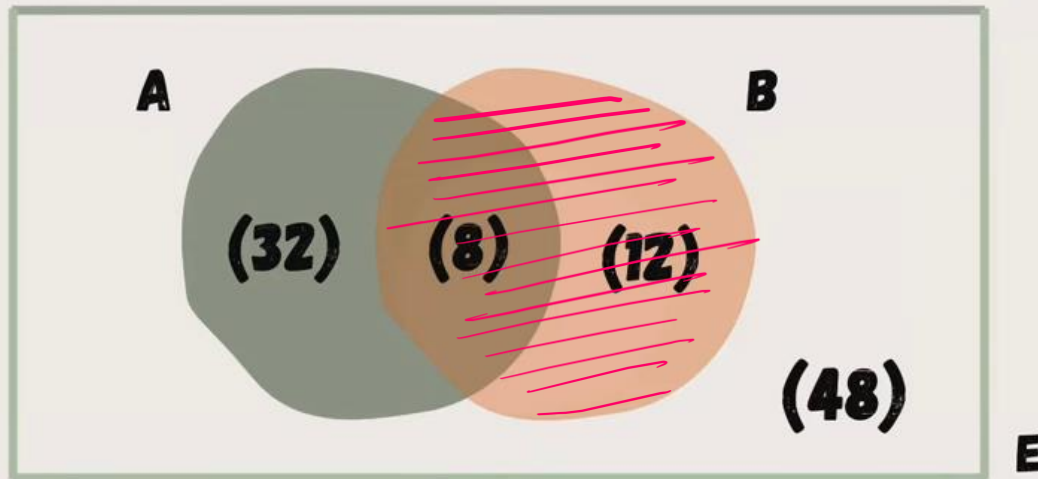
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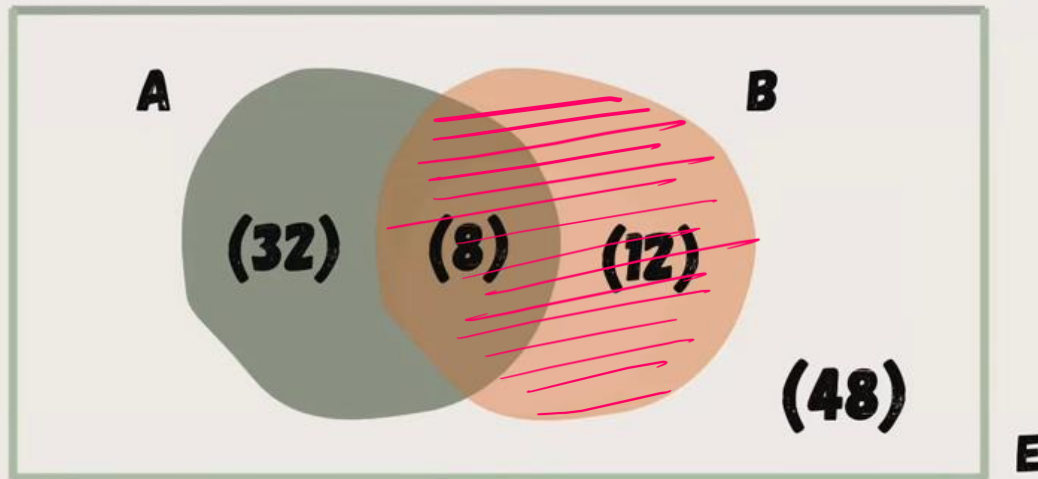
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$$P(A) = \frac{40}{100} = 0.4$$

$$P(A|B) = \frac{8}{20} = 0.4$$

same

$$= \frac{P(A \cap B)}{P(B)} = \frac{8/100}{20/100}$$

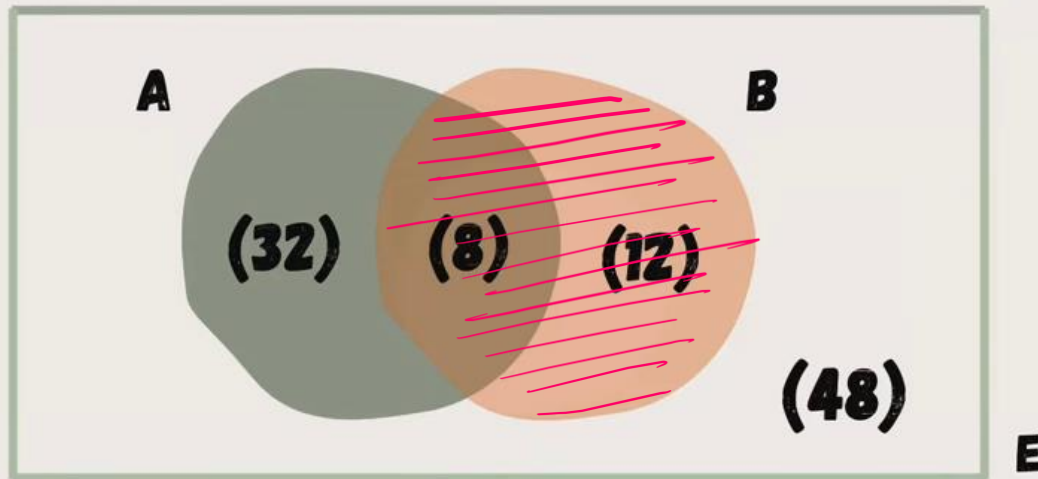
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$$P(A) = \frac{40}{100} = 0.4$$

$$P(A/B) = \frac{8}{20} = 0.4$$

same

$$= \frac{P(A \cap B)}{P(B)} = \frac{8/100}{20/100}$$

$$P(A/B)$$

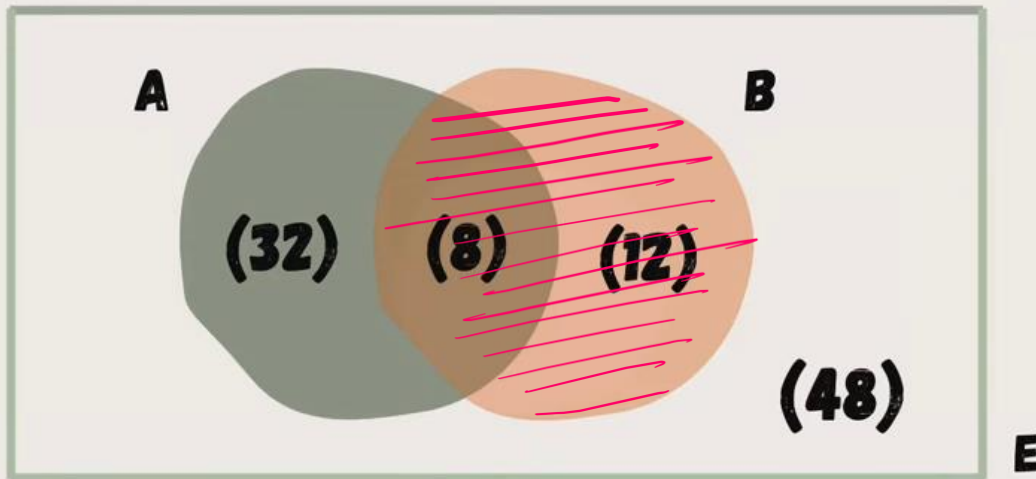
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$$= \frac{P(A \cap B)}{P(B)} = \frac{8/100}{20/100}$$

$$P(A) = P(A/B)$$

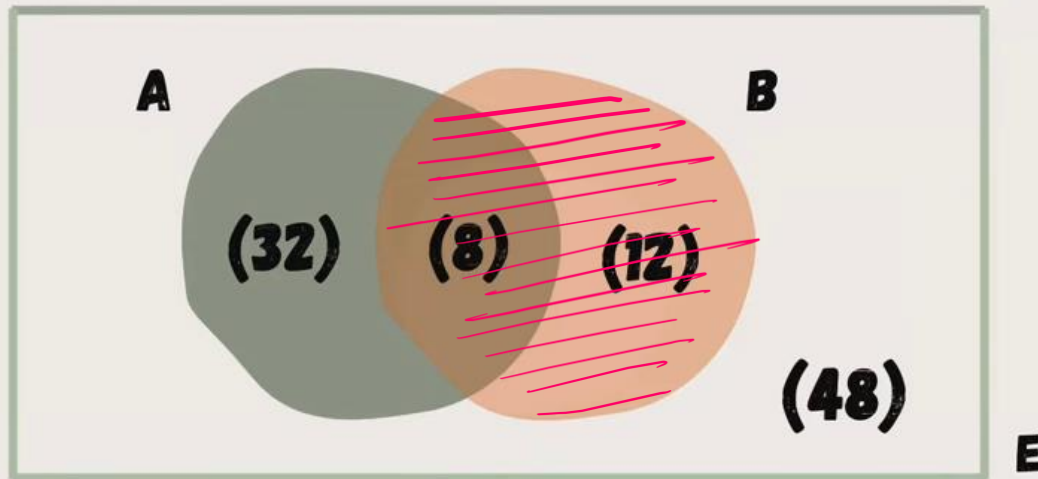
Independent Events

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same

$$= \frac{P(A \cap B)}{P(B)} = \frac{8/100}{20/100}$$

$$P(A) = P(A/B)$$

Similarly

$$P(B) = P(B/A)$$

Independent Events

Q: are these events independent?

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
A student plays
Basketball

EVENT B:
A student is taller
than 175 cm



Independent Events

Q: are these events independent?

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed

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Basketball

EVENT B:
A student is taller
than 175 cm



$$P(A) = \frac{40}{100} = 0.4$$

Independent Events

Q: are these events independent?

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
A student plays
Basketball

EVENT B:
A student is taller
than 175 cm



$$P(A) = \frac{40}{100} = 0.4$$

$$P(A|B) = \frac{35}{50} = 0.7$$

Independent Events

Q: are these events independent?

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
A student plays
Basketball

EVENT B:
A student is taller
than 175 cm



$$P(A) = \frac{40}{100} = 0.4$$

$$P(A|B) = \frac{35}{50} = 0.7$$

$$P(A) \neq P(A|B)$$

not independent

Independent Events

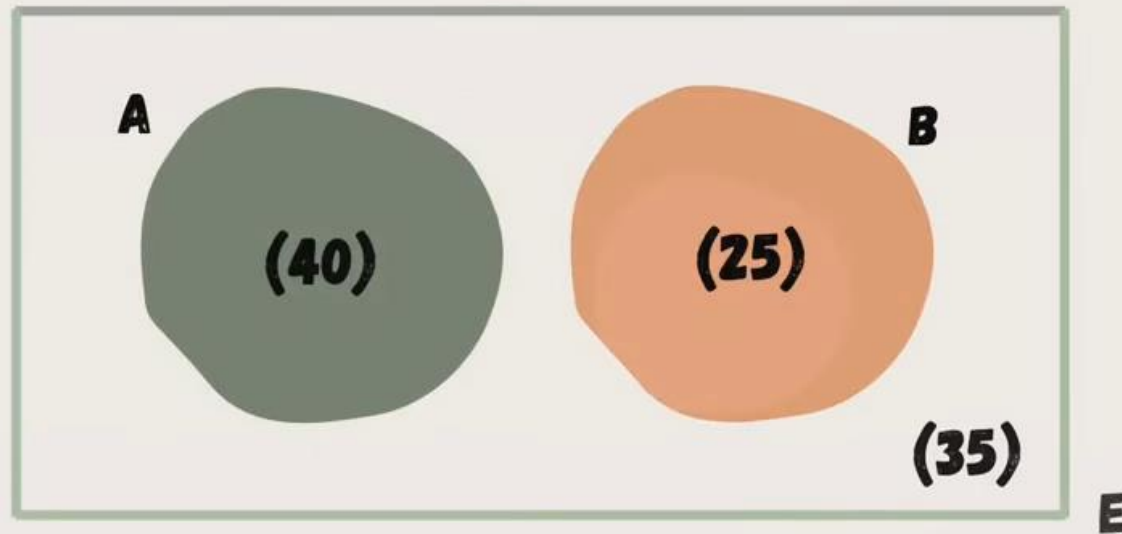
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MUTUALLY EXCLUSIVE EVENTS

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Independent Events

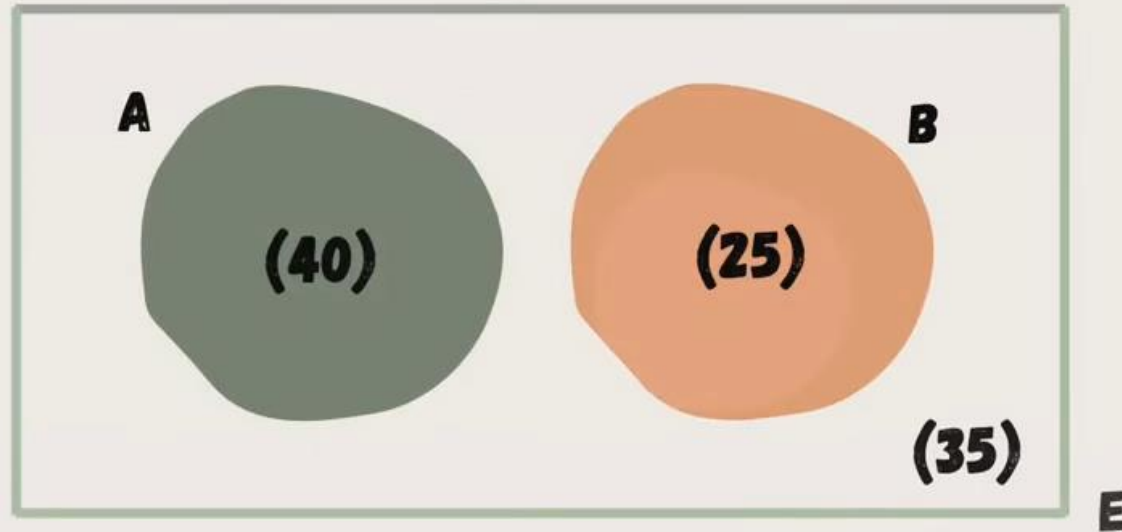
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Independent Events

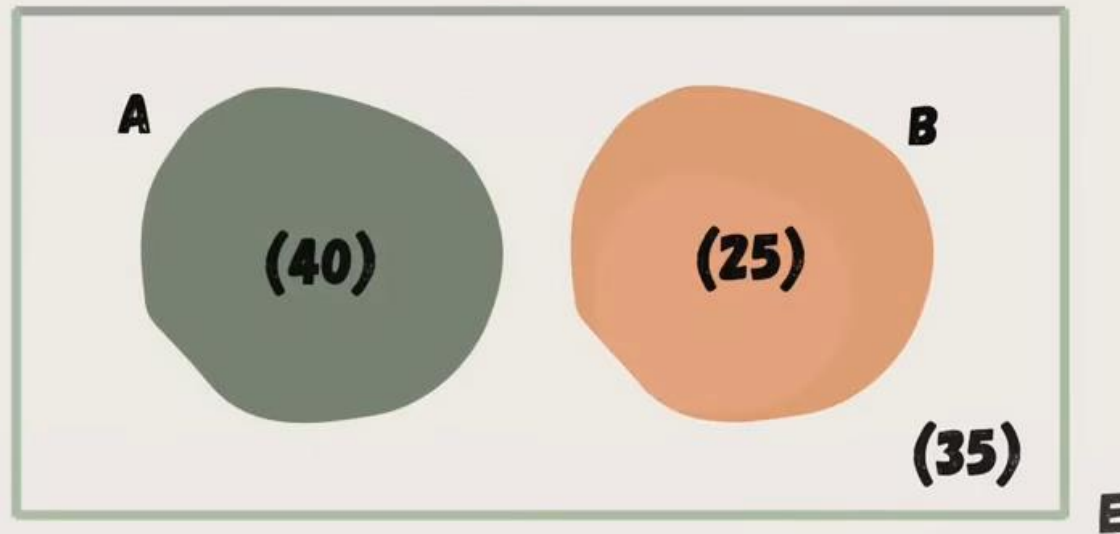
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$$P(A/B) = 0$$

Independent Events

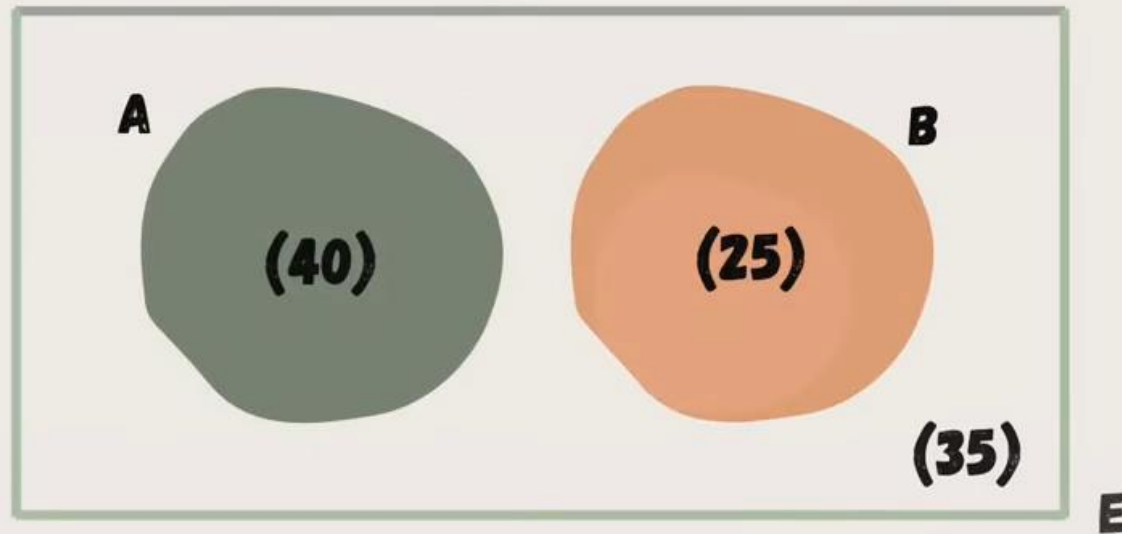
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$$P(A/B) = 0$$

not independent

Summary so far



Summary so far

$$P(A/B) = P(A)$$



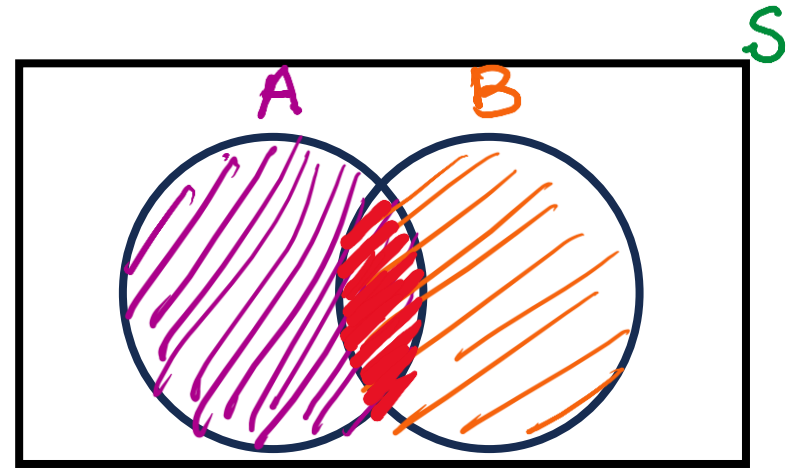
Summary so far

$$\begin{aligned} P(A/B) &= P(A) \\ P(B/A) &= P(B) \end{aligned} \quad \left. \vphantom{\begin{aligned} P(A/B) &= P(A) \\ P(B/A) &= P(B) \end{aligned}} \right\} \text{Independent Events}$$



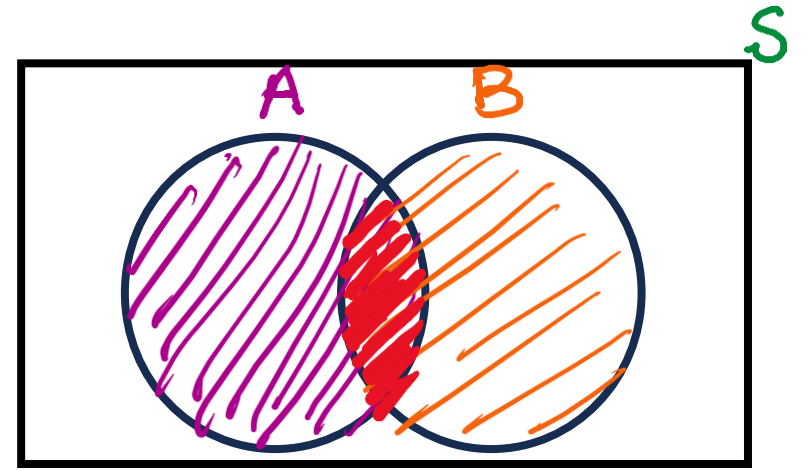
The product rule

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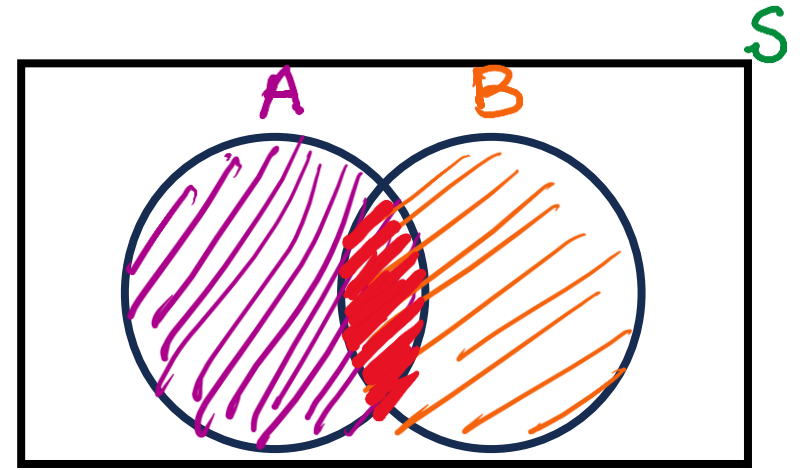
The product rule

$$P(A \cap B) = ?$$



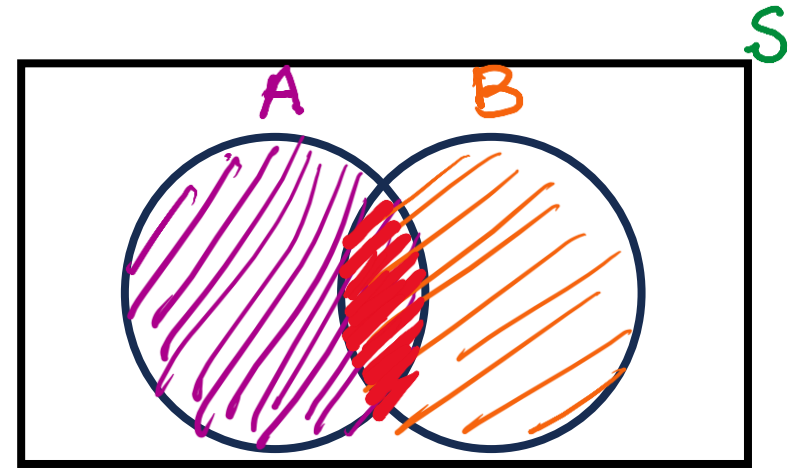
The product rule

$$P(A \cap B) = P(A|B)$$



The product rule

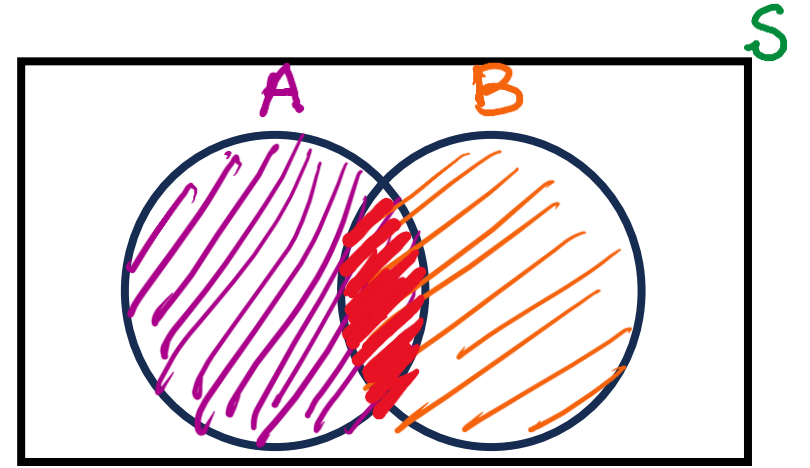
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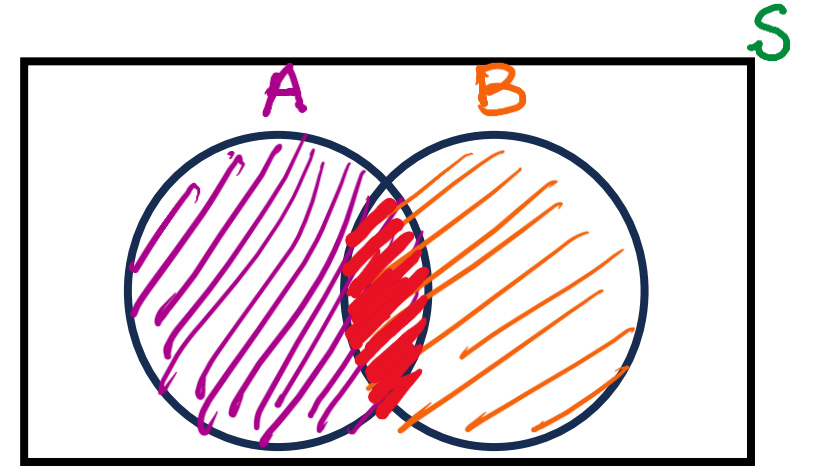
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The product rule

$$P(A \cap B) = P(B) P(A|B)$$

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Suppose that we have a fuse box containing 20 fuses, of which 5 are defective. If 2 fuses are selected at random and removed from the box in succession without replacing the first, what is the probability that both fuses are defective?



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$A \rightarrow$ first fuse is defective



Suppose that we have a fuse box containing 20 fuses, of **which 5 are defective**. If **2 fuses are selected at random** and removed from the box in succession without replacing the first, **what is the probability that both fuses are defective?**

A → first fuse is defective

B → second fuse is defective



Suppose that we have a fuse box containing 20 fuses, of **which 5 are defective**. If **2 fuses are selected at random** and removed from the box in succession without replacing the first, **what is the probability that both fuses are defective?**

$$P(A \cap B) = P(B) P(A|B)$$

$A \rightarrow$ first fuse is defective

$B \rightarrow$ second fuse is defective

$$P(A \cap B) = P(A) P(B|A)$$



Suppose that we have a fuse box containing 20 fuses, of **which 5 are defective**. If **2 fuses are selected at random** and removed from the box in succession without replacing the first, **what is the probability that both fuses are defective?**

$$\begin{aligned} P(A \cap B) &= P(B) P(A|B) \\ &= \frac{5}{20} \end{aligned}$$

$A \rightarrow$ first fuse is defective

$B \rightarrow$ second fuse is defective

$$P(A \cap B) = P(A) P(B|A)$$



Suppose that we have a fuse box containing 20 fuses, of **which 5 are defective**. If **2 fuses are selected at random** and removed from the box in succession without replacing the first, **what is the probability that both fuses are defective?**

$$\begin{aligned} P(A \cap B) &= P(B) P(A|B) \\ &= \frac{5}{20} \times \frac{4}{19} \end{aligned}$$

$A \rightarrow$ first fuse is defective

$B \rightarrow$ second fuse is defective

$$P(A \cap B) = P(A) P(B|A)$$



Suppose that we have a fuse box containing 20 fuses, of **which 5 are defective**. If **2 fuses are selected at random** and removed from the box in succession without replacing the first, **what is the probability that both fuses are defective?**

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$A \rightarrow$ first fuse is defective

$B \rightarrow$ second fuse is defective

$$\begin{aligned} P(A \cap B) &= P(A) P(B|A) \\ &= \frac{5}{20} \times \frac{4}{19} \quad \text{Ans} \end{aligned}$$



One bag contains 4 red balls and 3 black balls, and a second bag contains 3 red balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?



One bag contains **4 red balls** and **3 black balls**, and a second bag contains **3 red balls** and **5 black balls**. One ball is drawn from the first bag and placed unseen in the second bag. **What is the probability that a ball now drawn from the second bag is black?**

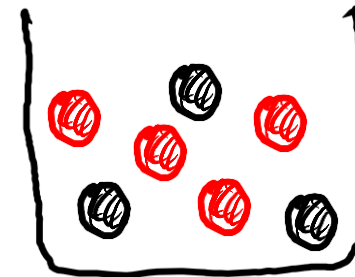


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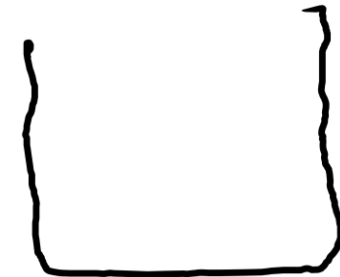




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Bag-1



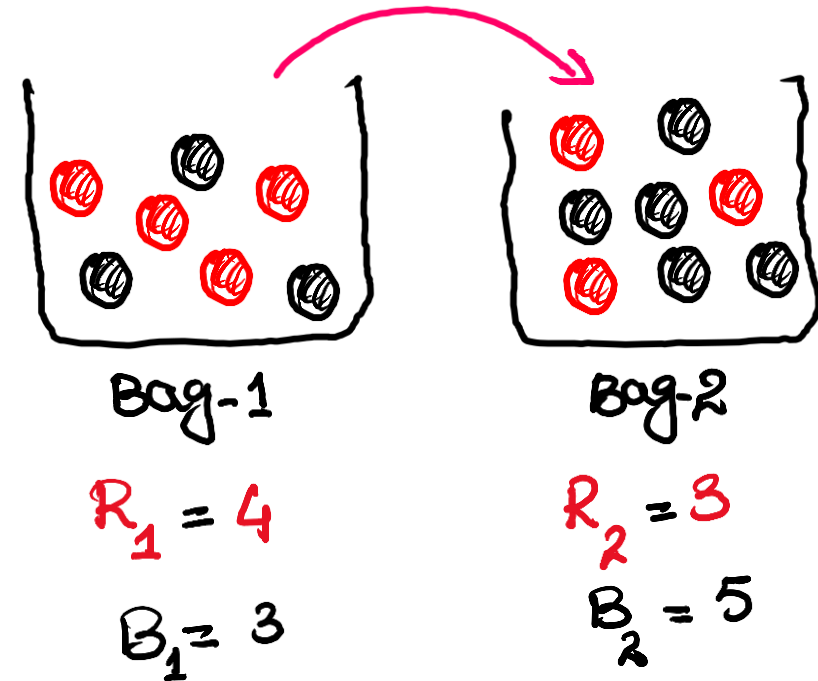
Bag-2

$$R_1 = 4$$

$$B_1 = 3$$



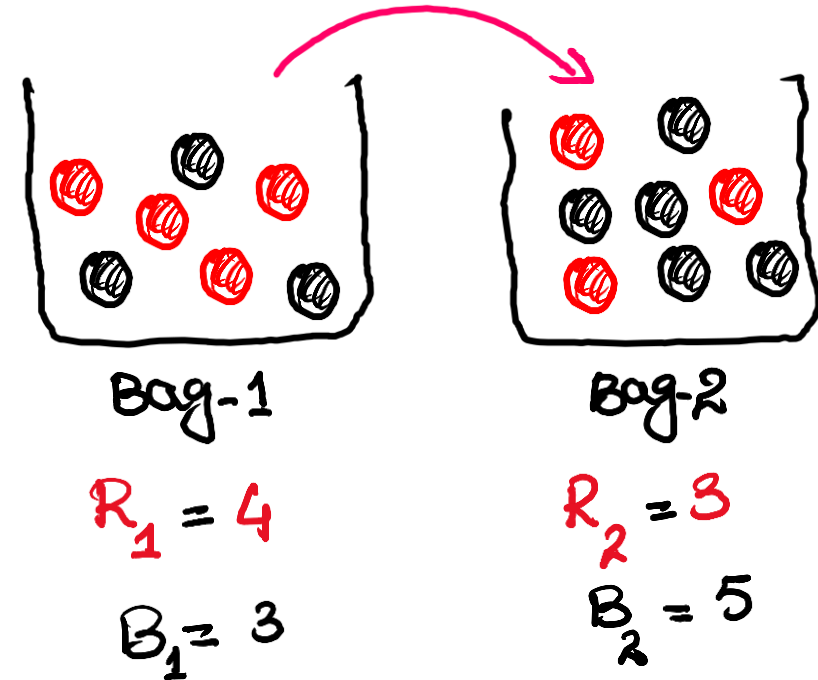
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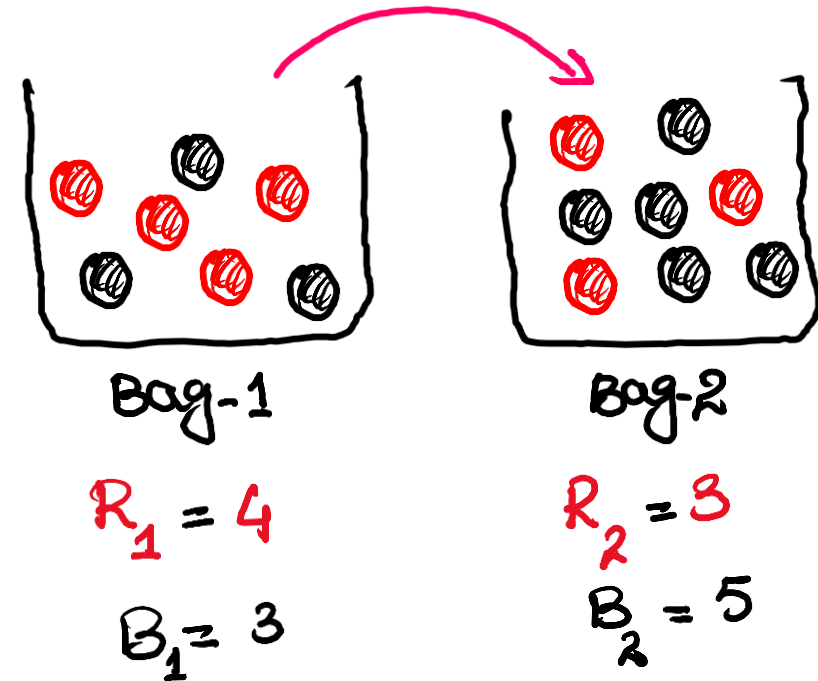
$$P[(B_1 \cap B_2)]$$





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$$P[(B_1 \cap B_2) \text{ or } (R_1 \cap B_2)]$$

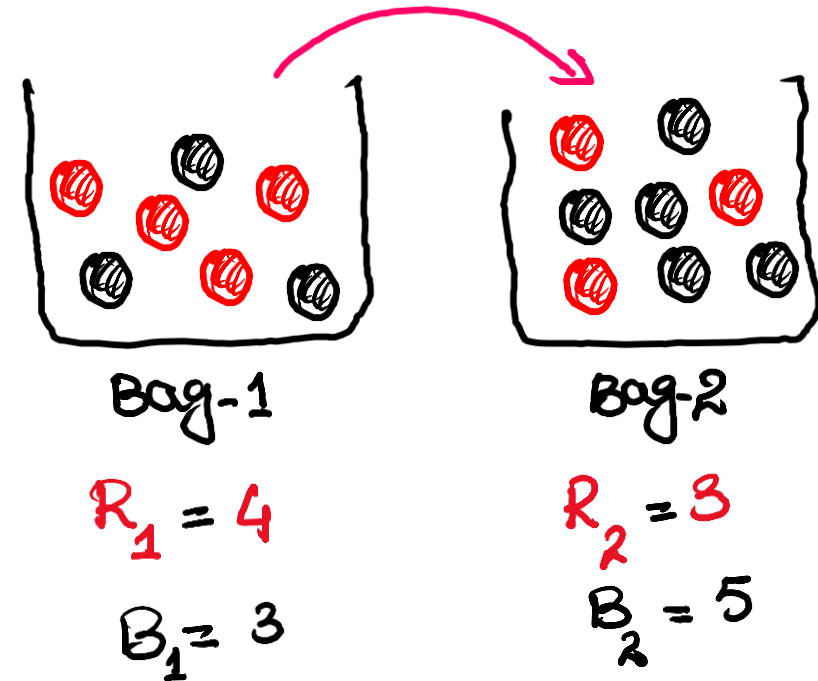




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$$P[(B_1 \cap B_2) \text{ or } (R_1 \cap B_2)] =$$

$$P[(B_1 \cap B_2) \cup (R_1 \cap B_2)] =$$





One bag contains **4 red balls** and **3 black balls**, and a second bag contains **3 red balls** and **5 black balls**. One ball is drawn from the first bag and placed unseen in the second bag. **What is the probability that a ball now drawn from the second bag is black?**

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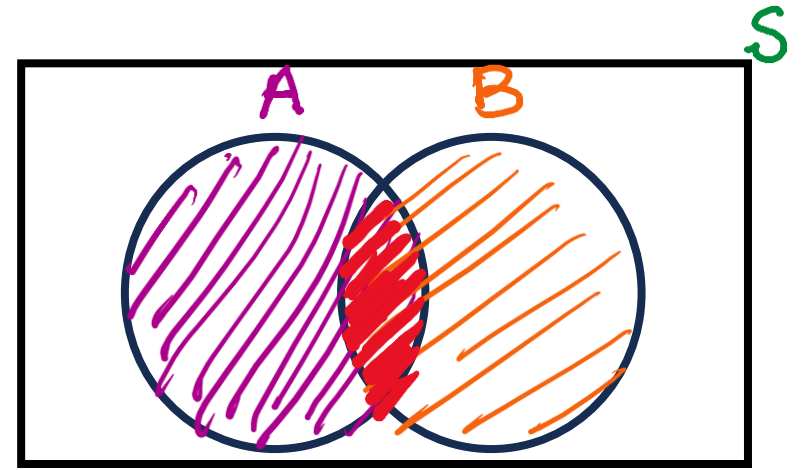
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The product rule

$$P(A \cap B) = P(B) P(A|B)$$

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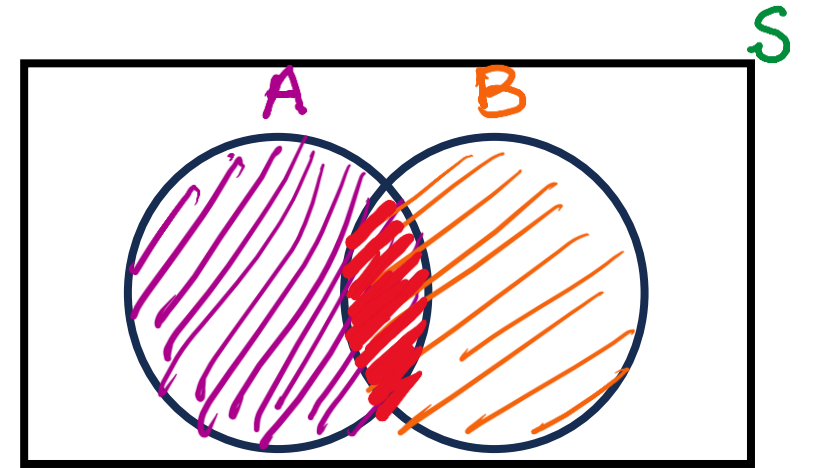


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For Independent Events



The product rule

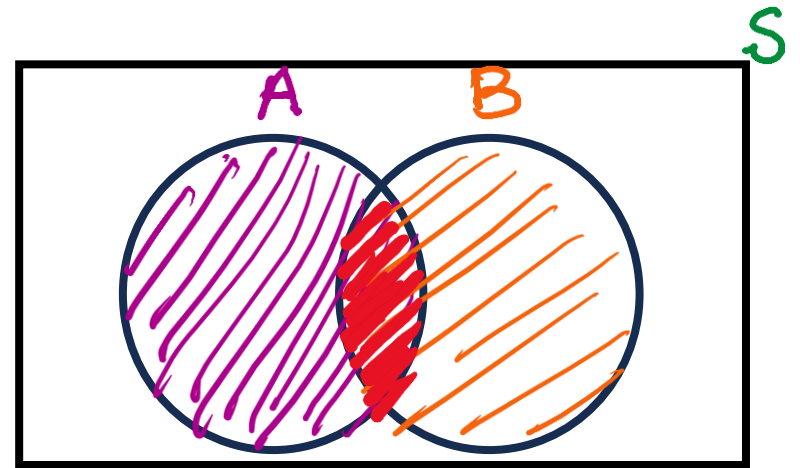
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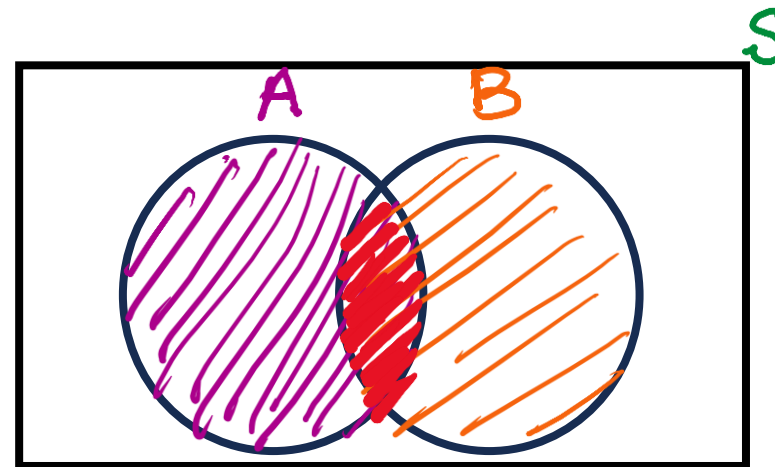
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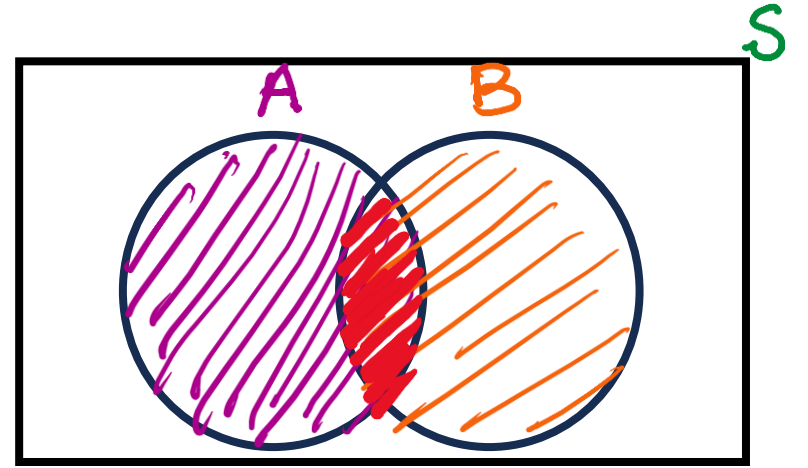
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A small town has one fire engine and one ambulance available for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and the fire engine will be available, assuming they operate independently.



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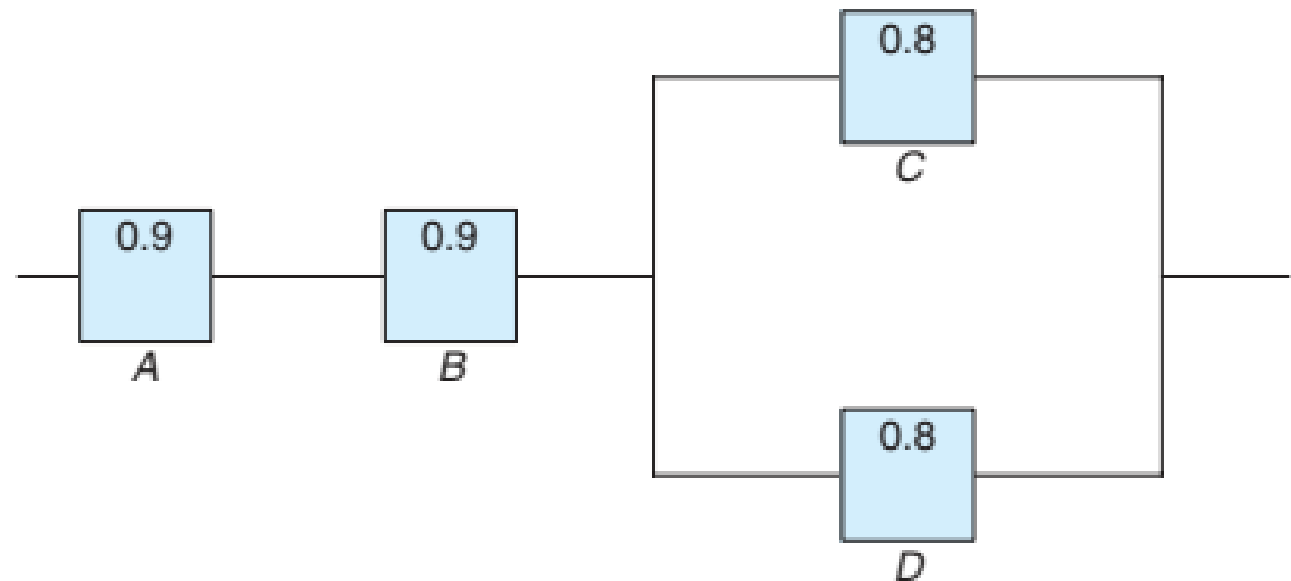


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$$\begin{aligned} P(A \cap B) &= P(A) P(B|A) = P(A) P(B) \\ &= 0.98 \times 0.92 \quad \text{Ans} \end{aligned}$$

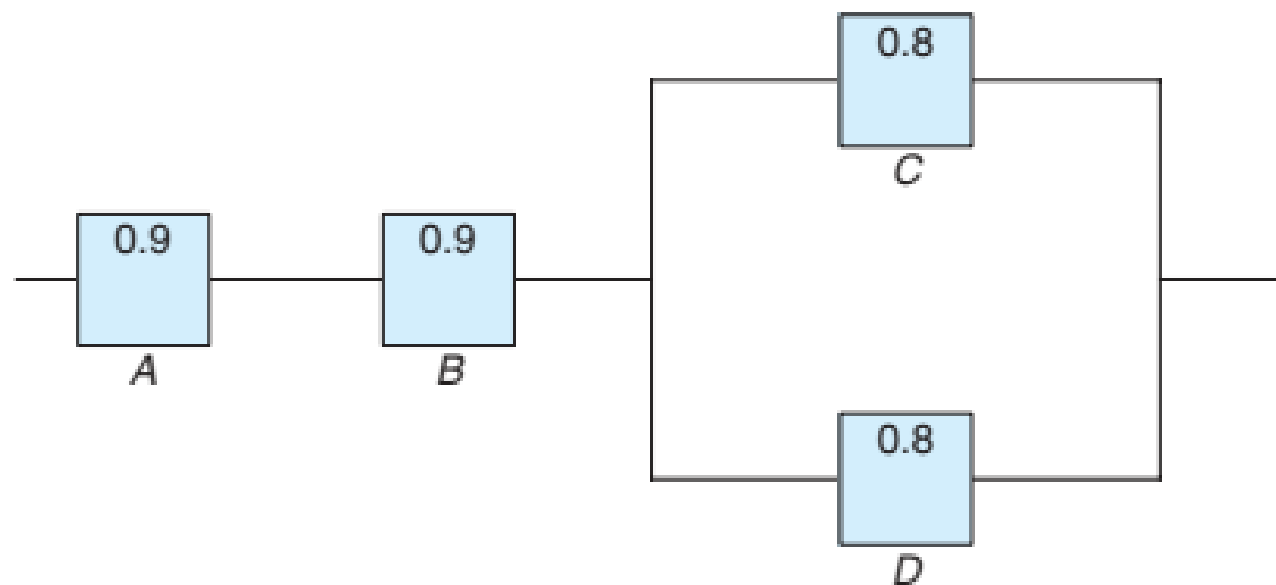


An electrical system consists of four components as illustrated in Figure. The system works if components A and B work and either of the components C or D works. The reliability (probability of working) of each component is also shown in Figure. Find the probability that (a) the entire system works and (b) the component C does not work, given that the entire system works. Assume that the four components work independently.





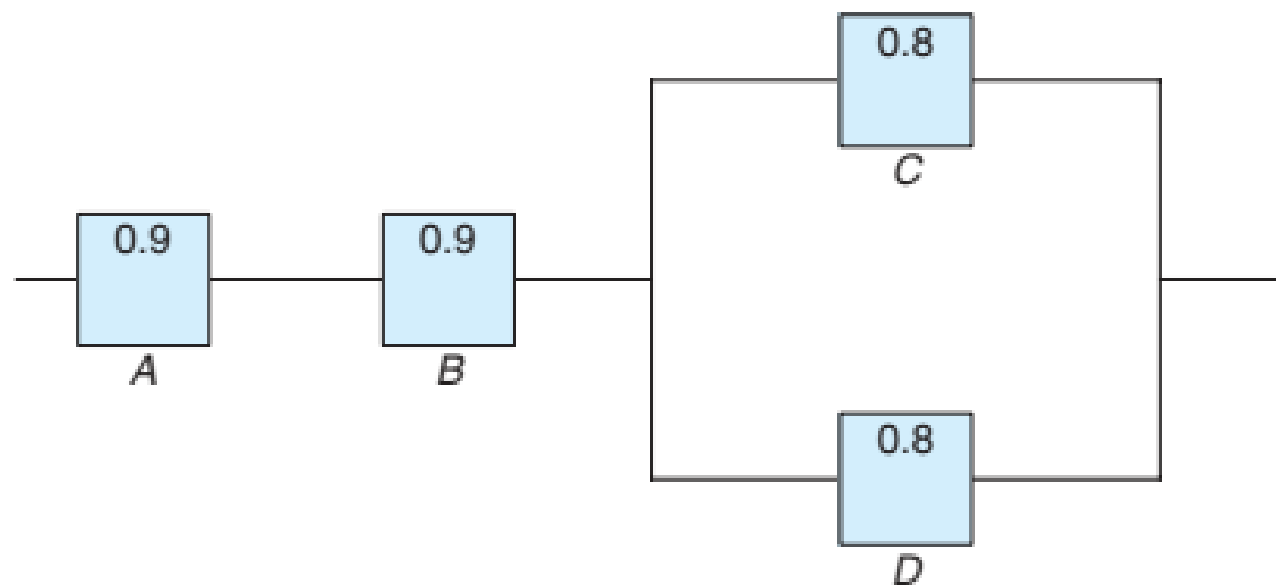
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The system works if **components A and B** work and **either** of the **components C or D works**. The probability of working of each component is shown in Figure. **Find the probability that (a) the entire system works.**

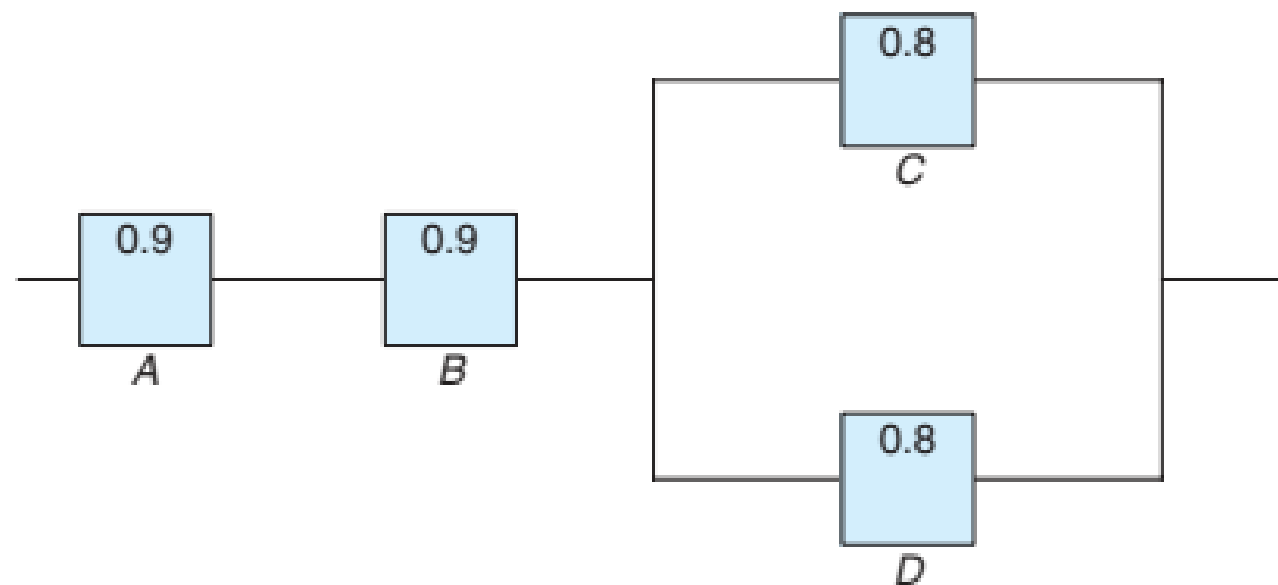
$$P(A \cap B)$$





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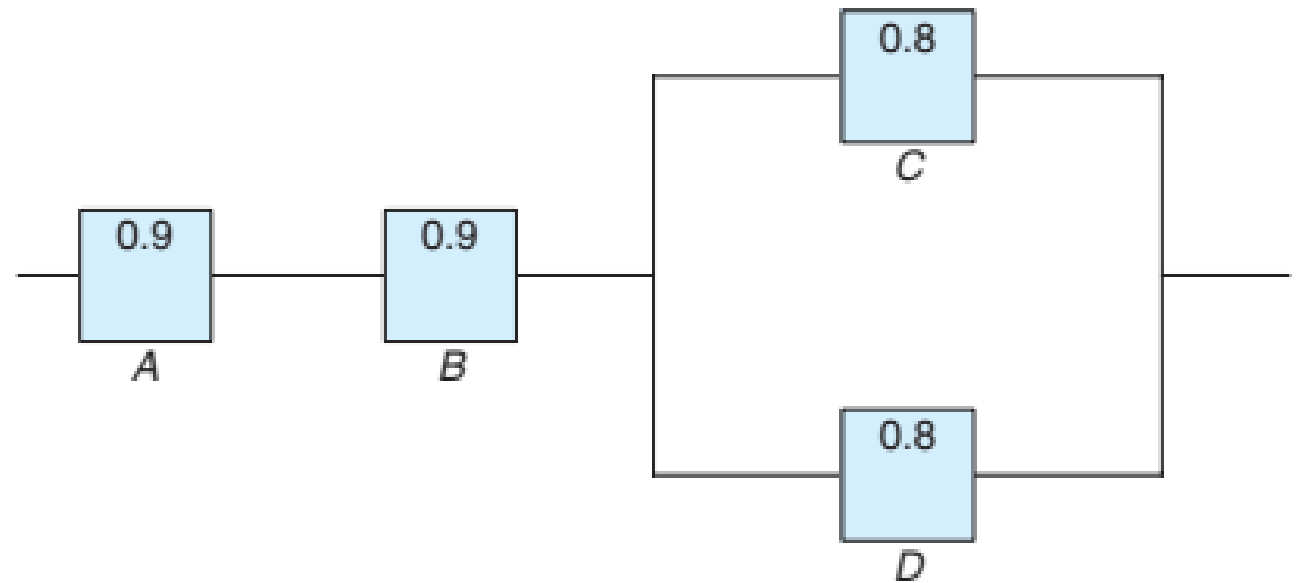
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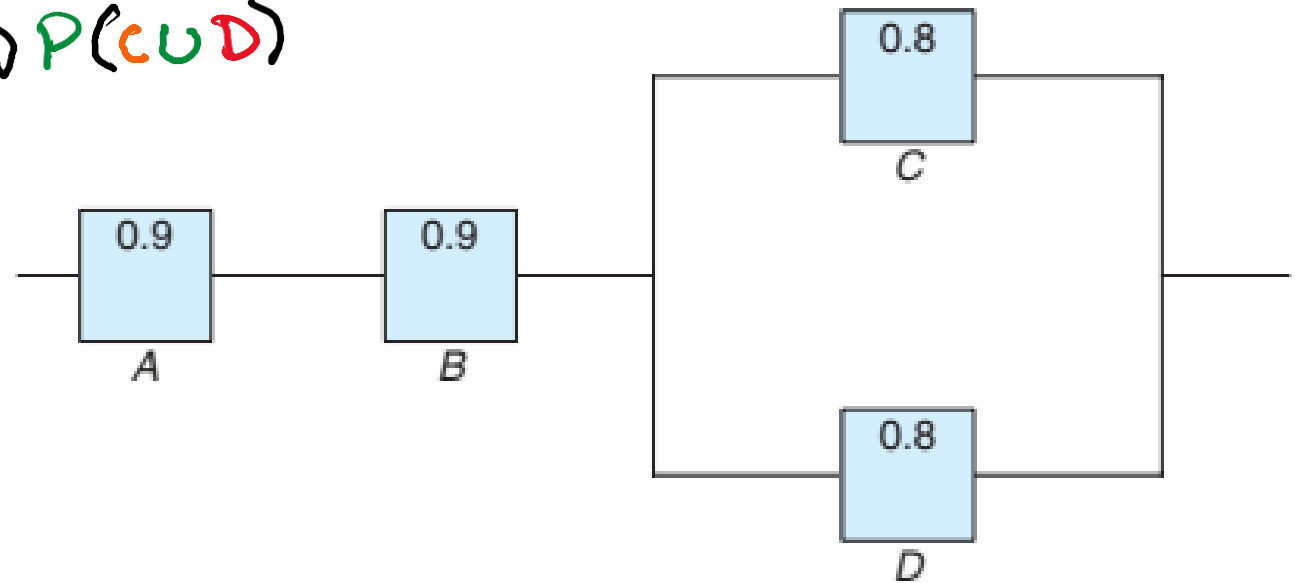
$$P(A \cap B \cap (C \cup D)) =$$





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$$P(A \cap B \cap (C \cup D)) = P(A) P(B) P(C \cup D)$$

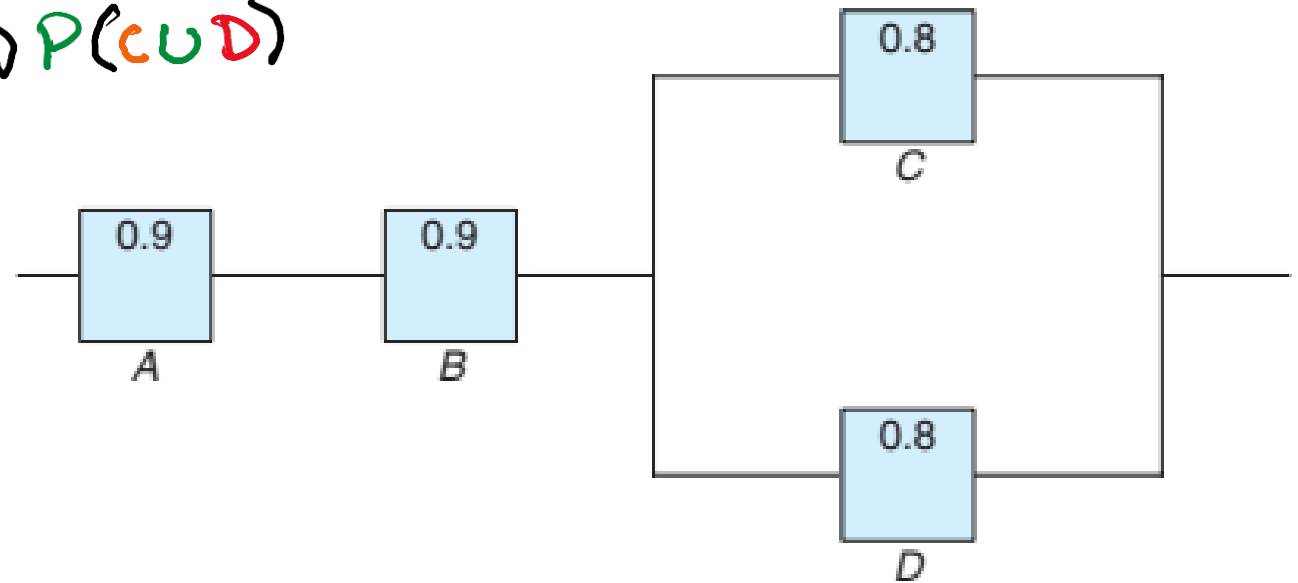




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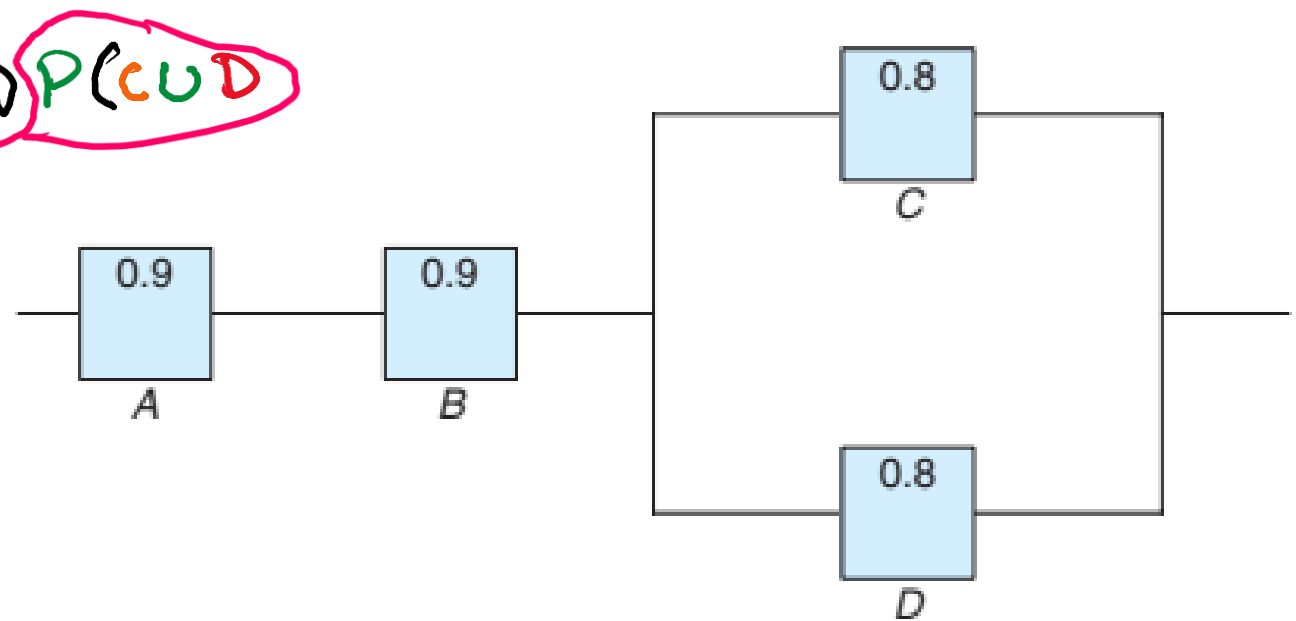
$$= P(A) P(B)$$





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$$P(A \cap B \cap (C \cup D)) = P(A) P(B) P(C \cup D)$$
$$= P(A) P(B) [1 - P(C' \cap D')]$$



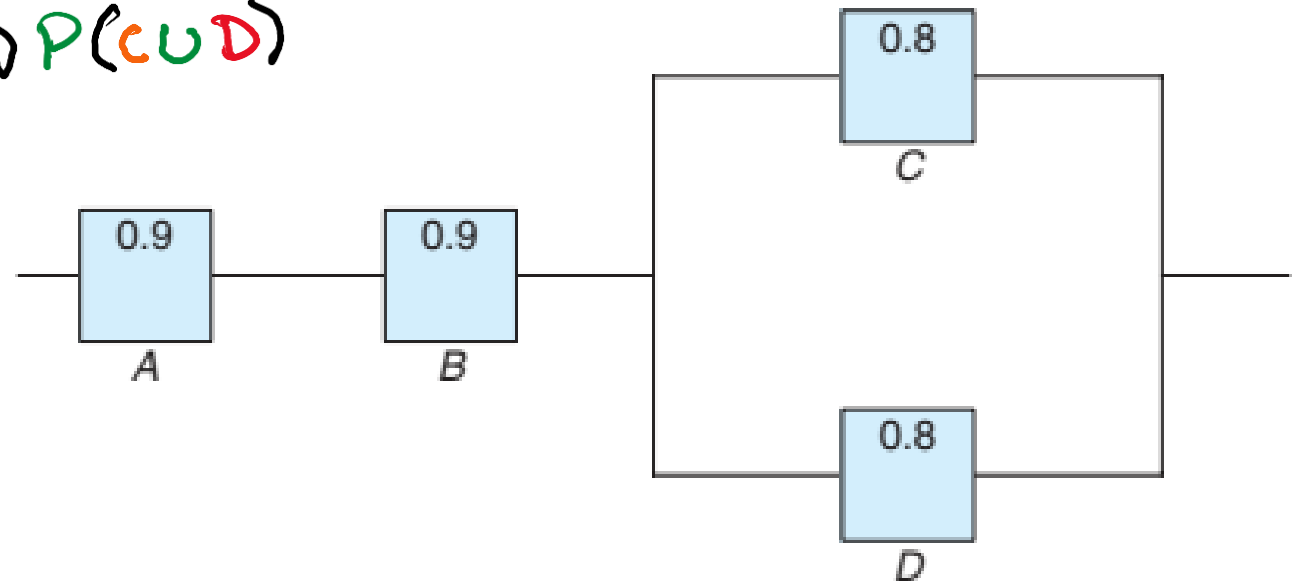


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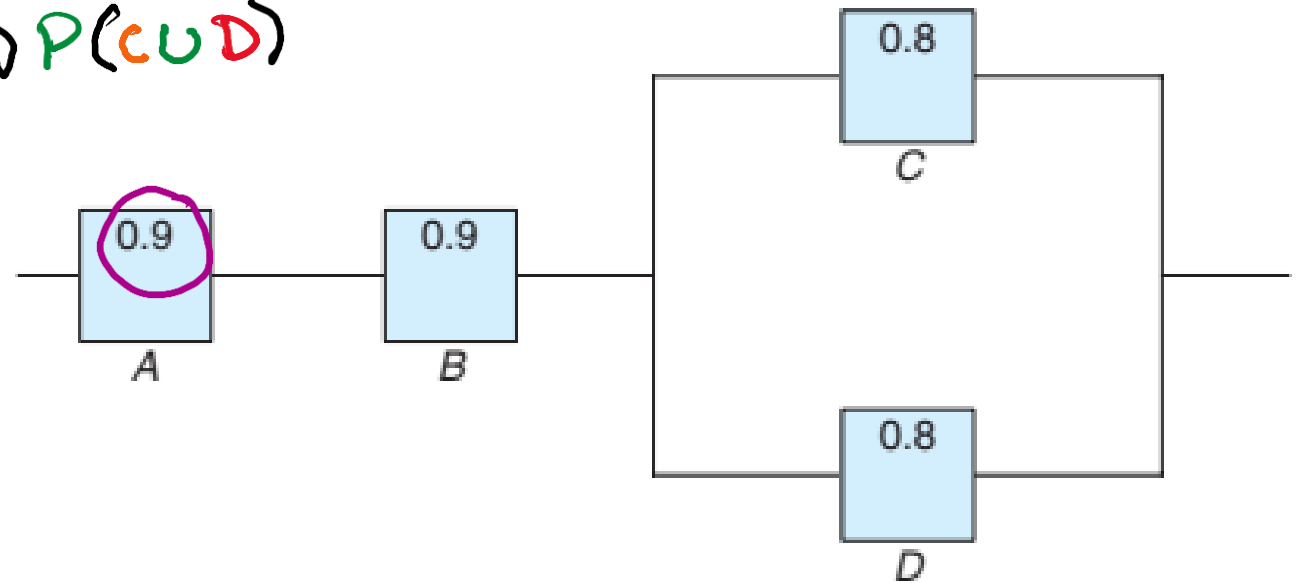
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$$= 0.9$$





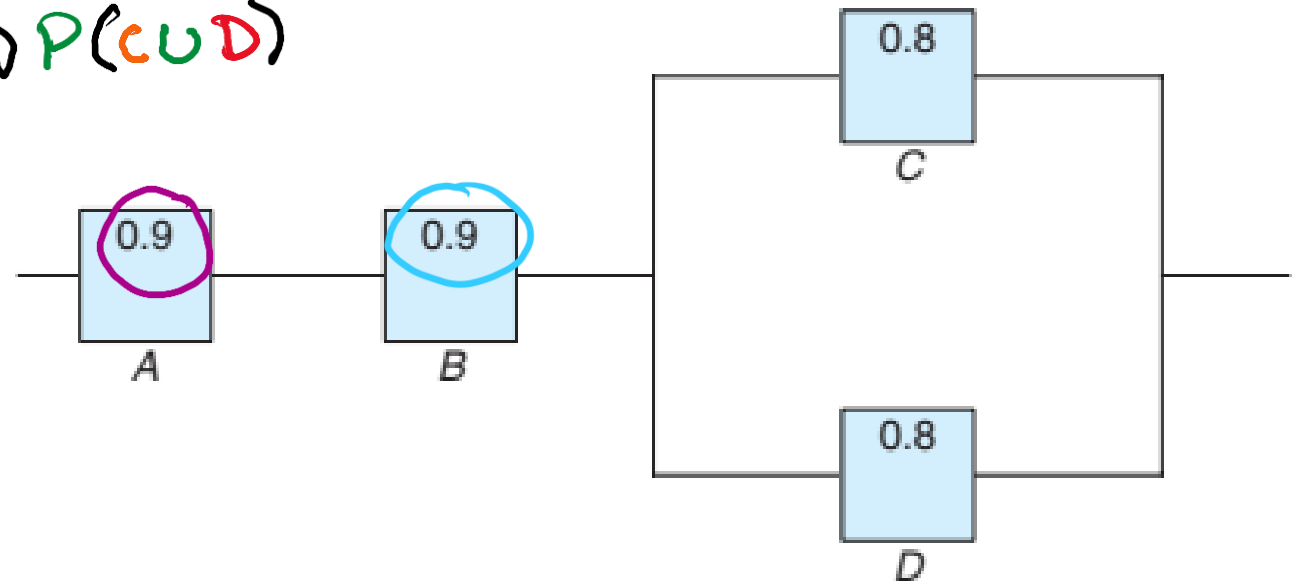
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$$= P(A) P(B) [1 - P(C') P(D')]$$

$$= 0.9 \times 0.9$$





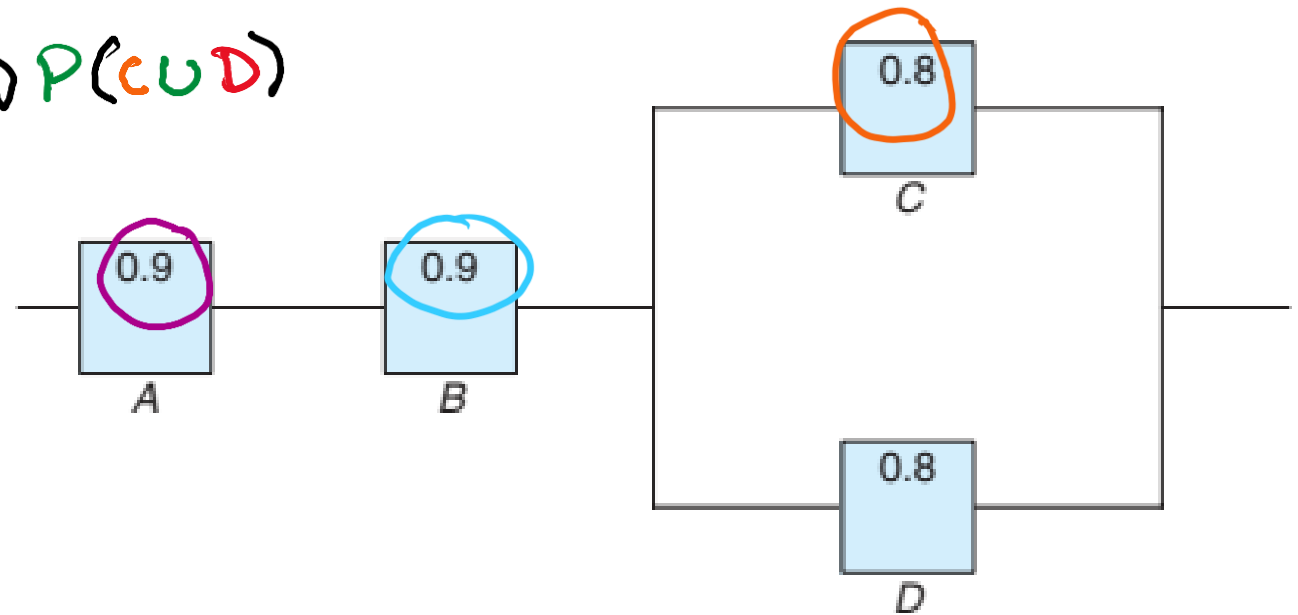
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$$= P(A) P(B) [1 - P(C') P(D')]$$

$$= 0.9 \times 0.9 \times [1 - (1 - 0.8)]$$





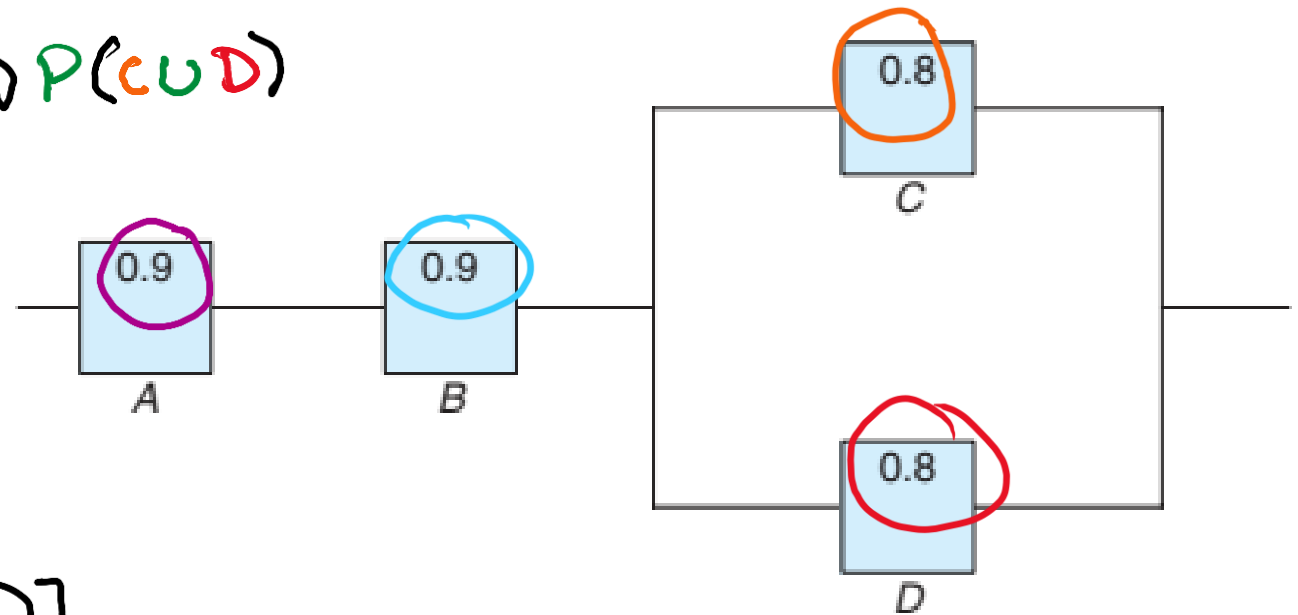
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$$= 0.9 \times 0.9 \times [1 - (1 - 0.8)(1 - 0.8)]$$

$$= 0.77 \quad \underline{\text{Ans}}$$

